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RESEARCH REPORT

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Psilocybin in alcohol use disorder and comorbid depressive symptoms: Results from a feasibility randomized clinical trial

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Abstract

Background and Aims: Psilocybin has emerged as a potential treatment for alcohol use disorder (AUD), but early efficacy data are inconsistent. Depression following alcohol detoxification significantly increases the risk of relapse. This pilot study aimed to evaluate the feasibility, acceptability, and preliminary efficacy of psilocybin-assisted psychotherapy for patients with comorbid AUD and depression.

Design: A prospective, single-center, double-blind, parallel (2:1), randomized controlled pilot study.

Setting: The study was conducted in a French inpatient addiction treatment program offering intensive relapse prevention interventions.

Participants: Of 350 screened patients, 30 adults (mean age 49 ± 10 years; 43% female) with severe AUD (*Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition [DSM-5] criteria) and a Beck Depression Inventory-II (BDI-II) score ≥14 were included. Participants had completed detoxification between 14 and 60 days prior to inclusion.

Interventions: Participants received either two oral sessions of 25 mg (n = 20) or 1 mg (n = 10) psilocybin-assisted psychotherapy spaced three weeks apart, as an add-on to standard care. Patients, investigators and outcome assessors were all blinded to patient group.

Measurements: The primary outcome was feasibility, according to participation in both dosing sessions and recruitment/inclusion rates. Secondary outcomes included alcohol use (Alcohol Timeline Followback), time to relapse, craving (Craving Experience Questionnaire), depression (BDI-II), safety and blinding integrity.

Findings: One participant in the 25 mg group could not receive the second dose due to myocardial infarction occurring three days earlier, unrelated to the treatment. Four participants in the control group refused the second session after guessing their group assignment (p-value = 0.019), with one participant self-administering 3,4-Methylenedioxymethamphetamine (MDMA). At 12 weeks, the 25 mg group showed significantly greater abstinent rate (11/20 (55%) vs 1/9 (11%) (one lost of follow up) (difference = −44%, [95% confidence interval [CI]: −82% to −5.9%]), p = 0.043), reductions in % drinking days −100 (−100 to −49) vs − 93 (−96 to 0), p = 0.038 and craving frequency −8 (−23

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to −1) vs + 7 (−2 to 11), p = 0.045, respectively in the 25 vs 1 mg groups (median [25;75]). Relapse rates were 35% in the 25 mg group and 50% in the control group (HR = 0.52 [0.16 to1.65]). No efficacy differences were observed based on antidepressant use in terms of drinking and depression. Blinding was imperfect (correct guess by patients: 93.3%; investigators: 86.7%). Twenty-five adverse events were reported in 10 patients (50%) in the 25 mg group versus 6 patients (60%) in the control group.

Conclusions: Psilocybin-assisted psychotherapy appears feasible, acceptable, and safe in recently detoxified patients with comorbid alcohol use disorder and depression.

KEYWORDS

alcohol use disorder, depression, dual disorders, feasibility, psilocybin, psychedelics, randomized controlled trial, relapse prevention

INTRODUCTION

Alcohol use disorder (AUD) is the fourth leading cause of preventable death, with an increasing prevalence in certain groups [1, 2]. It is a chronic disease, with an estimated 40% to 60% of individuals relapsing [3]. Recent data support a 'dual disorder' phenotype (i.e. the simultaneous presence of an addiction and another mental disorder) in individuals with AUD [4]. Fewer than 7% of patients with dual disorder receive treatment for both disorders [5], yet, treating both is key for rehabilitation and remission [6]. Moreover, individuals with dual disorders are often excluded from clinical trials [7].

Many individuals with AUD experience depression at some point. Depression is a risk factor for early relapse [8, 9], with approximately 60% of detoxified individuals with AUD plus depression relapsing within 3 months [10]. Comorbid depression increases sensitivity to negative reinforcement, and depression-induced changes in the reward system persist beyond the detoxification period, contributing to relapse [11]. Depression and craving are strongly correlated, persisting even after detoxification [12], with craving a main predictor of relapse [9].

The high prevalence of co-occurring AUD and depression, and their likely mechanistic overlap, justifies treatments targeting both [13]. Psilocybin, a psychedelic compound, has shown promising results in both depression and addictive disorders [14–16]. Psilocybin increases global brain connectivity, enhances brain entropy, plasticity, adaptability and the ability to change [17] and reduces feelings of social exclusion [18].

Efficacy data on psilocybin in AUD are limited and conflicting. An open-label, non-randomized trial on 10 non-detoxified patients [19] gave participants 0.3 mg/kg psilocybin, increased to 0.4 mg/kg 4 weeks later. Abstinence significantly increased following psilocybin and was largely maintained at 36-week follow-up. A larger randomized trial evaluated 25 mg/70 kg psilocybin versus diphenhydramine, with a higher dose in the second session in patients with alcohol use risk levels ranging from low to very high [20]. Psychotherapy included motivational enhancement therapy and cognitive-behavioral therapy. The percentage of heavy drinking days (HDD) during the 32-week study was significantly lower in the psilocybin group versus controls (9.7% vs. 23.6%). A more recent study suggested a possible efficacy of a single 25 mg dose at 12 weeks on heavy drinking [21], while another phase 2 trial on 37 recently withdrawn patients with mild-to-severe AUD found no significant difference on any drinking outcome after 6 months, using minimalist psychotherapy [22].

The efficacy of psilocybin in depression is more established [23–25]. A meta-analysis including seven trials in 436 adults with depression [26], found an overall effect size of 0.66 (95% CI = 0.46–0.86). Recent data suggest that psychedelics could serve as add-on treatment to antidepressants [27]. Resistance to antidepressants has been repeatedly demonstrated to be linked to alcohol consumption [28, 29]. No data are available for efficacy of psilocybin in dual disorders.

Depression must be treated soon after alcohol detoxification to prevent early relapse, however, approved antidepressant treatments are slow acting [30]. Psilocybin's immediate therapeutic effects make it a promising medication for this dual disorder population.

We hypothesized that two oral administrations of 25 mg psilocybin, 3 weeks apart, compared to very low dose psilocybin (1 mg) would be acceptable in recently detoxified in-patients with AUD and persistent depressive symptoms. This pilot study aimed to evaluate the feasibility, acceptability and preliminary efficacy outcomes of psilocybin for post-detoxification AUD with depression, alongside usual care [31].

METHODS

Design

This was a prospective, single-center, double-blind, parallel, randomized controlled pilot study (2:1) comparing two oral administrations of psilocybin-assisted therapy three weeks apart of 25 mg psilocybin (experimental group) versus a very low dose of 1 mg psilocybin (control group), as an add-on to treatment as usual, including an intensive relapse prevention program, in alcohol use disorder and persistent depressive symptoms after detoxification, on drinking and depression with a 12-week follow-up (Figure 1). This study was conducted in accordance with the principles of the Declaration of Helsinki and adhered to Good Clinical Practice guidelines. It was approved by the French Ethics Committee (Comité de Protection des Personnes) and the French National Agency for the Safety of Medicines and Health Products (2023–506 647–40-02) and registered on clinicaltrials.gov (NCT06235411). All participants provided written informed consent.

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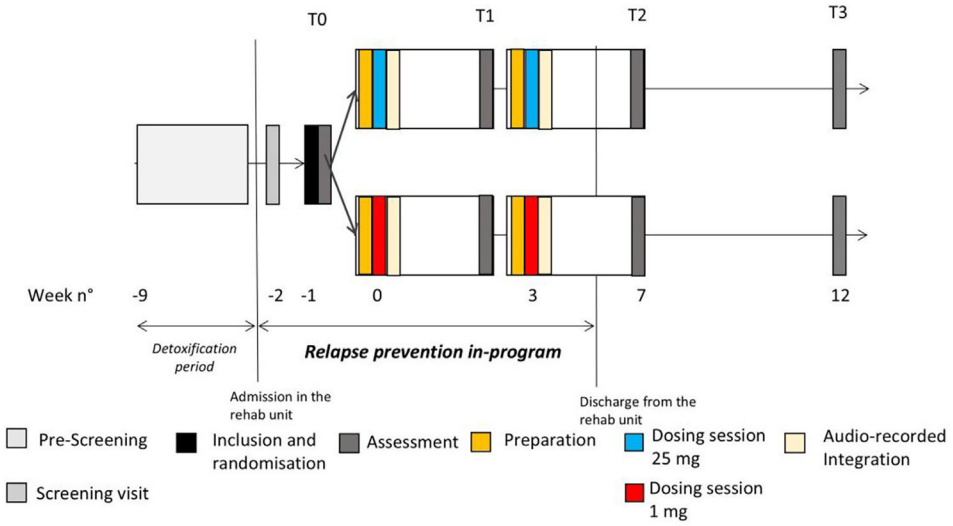


FIGURE 1 Study design.

Participants

Patients >18 were eligible if they had severe AUD confirmed by Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) criteria (i.e. 6 or more) and a Beck Depression Inventory-II (BDI-II) score ≥14. Their last alcohol consumption had to have occurred between 60 and 14 days before inclusion, with at least one HDD during their last 4-week drinking period.

Exclusion criteria included any history of schizophrenia or psychotic disorders, past or current manic or hypomanic episodes or the need for antipsychotic or monoamine oxidase inhibitor. Patients with current suicidal ideation classified as 'high risk' on the Columbia-Suicide Severity Rating Scale (C-SSRS) or those with a first-degree family history of psychotic or bipolar type 1 disorder were excluded. Patients deemed at high risk for negative emotional or behavioral reactions, including those with severe personality disorders, or lack of social support, were also ineligible. Other exclusion criteria included dementia or severe cognitive impairment, a Clinical Institute Withdrawal Assessment for Alcohol-Revised (CIWA-R) score ≥8 and medical conditions that could compromise safety, such as epilepsy, significant liver dysfunction, coronary artery disease, arrhythmia, heart failure, uncontrolled hypertension (≥165/95 mm Hg), history of stroke, severe asthma, hyperthyroidism, narrow-angle glaucoma, severe gastrointestinal obstruction, symptomatic prostatic hypertrophy or uncontrolled diabetes. Individuals with a history of hallucinogen use disorder, any hallucinogen use in the past year or more than 25 lifetime uses were excluded, as well as those with cocaine, psychostimulant, opioid or cannabis dependence in the past 12 months.

Severe electrocardiography (ECG) abnormalities such as signs of ischemia, myocardial infarction or prolonged QTc (>0.45 seconds in men, >0.47 seconds in women) led to ineligibility. Practical constraints included lack of email access, insufficient proficiency in French or concurrent participation in another drug trial. Pregnant or breastfeeding women and those planning a pregnancy were ineligible.

Procedures

Patients were hospitalized in a specialized rehabilitation center (CHU de Nîmes) for 4 weeks following detoxification and discharged the day after the second dosing session. Treatment as usual comprised a comprehensive 4-week relapse prevention program including cognitive behavioral therapy (CBT) focused on relapse prevention, motivational interviewing, group cognitive remediation, physical exercise, mind–body psychological support and psychomotricity care. Anti-craving medications and conventional antidepressants were authorized. In-patients were pre-screened on arrival and those detoxified elsewhere and scheduled for hospitalization at the recruiting center were phoned in advance to inform them about the study. The study received high media exposure, leading to proactive outreach from potential participants. Those who contacted the research team were pre-screened by phone by a psychiatrist investigator and were sent the information sheet to allow sufficient time for consideration. Eligible participants were included within 14 to 60 days of abstinence, depending on availability of psilocybin sessions. At a screening visit up to 7 days before inclusion, eligibility was rechecked and patients received information regarding the study. The first dosing session was scheduled within 7 days after inclusion. Each dosing session was preceded the day before by a preparation session, and followed the day after by an integration session. Therapists involved in the three phases—preparation, dosing session and integration—were psychiatrists or psychologists with experience in addiction and

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trauma, as well as in psychotherapeutic approaches involving altered states of consciousness (e.g. eye movement desensitization and reprocessing, hypnosis and mindfulness). All therapists adhered strictly to the study's therapeutic manual (see Supporting information Data S1). The second dosing session occurred 21 days later (± 3 days). Follow-up assessments were conducted on-site at 3 weeks, by phone by a research assistant independent of the care team or on-line for questionnaires.

Randomization

Patients were randomized 2:1 with a ratio in favor of the experimental arm, and stratified by gender. The randomization sequence was generated using SAS software v9.4 at the Nîmes University Hospital, with random permutation of block sizes 3 and 6 by a statistician independent of the study. Randomization was then implemented through the Inclusio software.

Interventions

The intervention was strictly the same in both groups except for the dosage (1 vs. 25 mg).

Preparation

The preparation session aimed to establish a safe and structured environment, introduce the psychotherapeutic process and help participants define positive intentions to guide the psychedelic session. Information was provided on the concepts of 'set and setting', with an emphasis on physical comfort, the role of music and the structure of the session, followed by an explanation of psilocybin's mechanisms (serotonergic action, neuroplasticity and potential anti-inflammatory effects). Misconceptions about psychedelic experiences were addressed, including variability in effects and emotional intensity. A formal mindfulness exercise with two focal points was conducted before intention elicitation to enhance present-moment awareness. The acceptance and commitment therapy (ACT) approach was used to help patients formulate a personally meaningful intention, focusing on values and desired changes rather than avoidance goals. Techniques for emotional grounding were introduced (breathing exercises, tactile anchoring, recalling self-identity and body scanning). Consent for reassuring touch of hand and arm during the session was collected [32]. Patients were informed about temporary psychological and physical effects, emotional processing and safety measures (continuous monitoring, emergency support and rescue medication). The most common potentially serotonergic medications were prohibited the day before and the day of dosing to avoid serotonin syndrome. Additionally, monoamine oxidase inhibitors were prohibited.

Dosing sessions

On the morning of the psilocybin sessions, patients were required to be afebrile, normotensive and not tachycardic. Dosing took place in the patient's room (Figure S1), with their personal belongings present and music facilitating the psychedelic experience playing [33]. Patients were instructed to lie down and wear an eye mask. A psychologist or psychiatrist trained in assisting experimental psychedelic sessions remained in the room throughout the 6-hour duration. The support provided followed a non-specific supportive psychotherapy model, using an affirmative yet largely non-directive approach, encouragement in facing inner experiences (including difficult thoughts, sensations or memories) and emotional support. Vital sign monitoring (blood pressure, heart rate) was conducted every 30 minutes for the first 2 hours, then hourly. During these checks, participants orally completed three 0 to 10 rating scales: intensity of the experience; anxiety levels; and perceived threat.

Integration

The following day, a 1-hour integration session was conducted by a therapist and audio-recorded. Results from these recordings will be reported separately. A semi-structured interview explored: (a) were the patient's expectations for the session met? (b) Was there any unexpected effect? (c) A summarized narrative of the psychedelic experience during the session; (d) Did the patient encounter any difficulties? and (e) What was the perceived usefulness of the experience in relation to their AUD, mood and life situation? The therapist helped the patient to consolidate the insights gained and initiate behavioral changes to support remission from AUD and depressive symptoms. Any psychiatric emergencies or potential risks, whether related to psilocybin administration or not, were assessed and managed by the investigating psychiatrists throughout the study.

Measures

The primary outcome was feasibility, assessed as the number of patients completing both psilocybin sessions, and screening and inclusion rates. Acceptability was measured by the number of patients who prematurely withdrew from the study, with reasons.

Efficacy outcomes focused on alcohol consumption patterns, assessed using the last 4-week Alcohol Timeline Followback (TLFB) method at 6 and 12 weeks after inclusion [34, 35], evaluating changes in the number of heavy drinking days (HDDs)—i.e., days with consumption of ≥6 standard drinks (10 g of alcohol), corresponding to the World Health Organization definition of heavy episodic drinking [36]—as well as the number of drinking days, total alcohol intake, and both the percentage of participants who relapsed and time to relapse at 12 weeks. Relapse was defined as at least one HDD. Abstinence was defined as no drinking in the past month. Additional assessments included:

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- Craving Experience Questionnaire (CEQ) for craving intensity and frequency [37]. The CEQ retrospectively assesses, over a defined time period, the intensity (S, strength) and frequency (F, frequency) of craving—defined as an irresistible urge to consume a substance or food. The CEQ can be used to evaluate craving for a variety of substances, including alcohol, cigarettes, drugs or food. It demonstrates excellent psychometric properties. The CEQ allows for the calculation of a craving intensity score for a given substance or food, based on 11 intensity items from sections 'a', 'b' and 'c.' Each item is rated from 0 ('Not at all') to 10 ('Extremely'). The sum of these 11 ratings yields a total craving intensity score ranging from 0 to 110. The higher the score, the more intense the participant's reported craving. A second score provided by the CEQ reflects the frequency of craving. It is calculated by summing the values of the 11 items from sections 'd', 'e' and 'f.' Each item in this section is rated from 0 ('Never') to 10 ('Constantly'). The total craving frequency score also ranges from 0 to 110.
- Alcohol Quality of Life Scale (AQoLS) for alcohol-related quality of life [38,39]. The AQoLS is an instrument designed to assess health-related quality of life (HRQoL) from the patient's perspective in individuals with AUD and it has demonstrated good psychometric properties. It is the only HRQoL instrument specifically developed for people with alcohol-related problems, and was created based on direct patient input, allowing for a more accurate and relevant assessment of HRQoL in this population. The dimensions of control and self-esteem are particularly valuable, as they are not captured in generic HRQoL tools used for individuals with attention disorders. The AQoLS can be used in clinical research to evaluate the effectiveness of therapeutic interventions regardless of drinking goals, providing insights into their relevance from the patient's viewpoint. It consists of 34 items assessing the negative impact of alcohol use on quality of life across 7 dimensions: social relationships, activities, living conditions, self-care, negative emotions, sleep and loss of control. Each item is rated on a 0 ('Not at all') to 3 ('Extremely') scale, with the total score being the sum of all items, ranging from 0 to 102. There is no threshold value; the patient serves as their own reference point over time. The scale can be repeated every 4 weeks.
- BDI-II for depressive symptom severity [40]. We present two exploratory *post hoc* outcomes: at least 50% [41] and 80% reduction in BDI score at week 12. The BDI-II is a 21-item self-report scale used to assess the severity of depressive symptoms. Participants are instructed to circle the number corresponding to their chosen statement and may select multiple responses per item if more than one applies. Each item consists of four statements reflecting increasing levels of symptom severity, scored from 0 to 3. For scoring, only the highest selected score for each item is retained. The total score is the sum of the 21 item scores, ranging from 0 to 63, with the following cut-offs: 0 to 13 (minimal), 14 to 19 (mild), 20 to 28 (moderate) and 29 to 63 (severe depression).
- Beck Anxiety Inventory (BAI) for anxiety severity [42]. The BAI is a 21-item multiple-choice self-report inventory used to assess the severity of anxiety. It measures common anxiety symptoms experienced over the past week, including the day of assessment. Each item is scored from 0 ('Not at all') to 3 ('Severely'), with higher total scores indicating more severe anxiety symptoms. The total score is the sum of the 21 item scores, ranging from 0 to 63, with the following cut-offs: 0 to 7 (minimal), 8 to 15 (mild), 16 to 25 (moderate) and 26 to 63 (severe).
- Difficulties in Emotion Regulation Scale (DERS) for emotion regulation difficulties [43]. The DERS is a brief 36-item self-report questionnaire designed to assess various dimensions of emotion dysregulation. It yields a total score as well as six subscale scores derived from factor analysis: non-acceptance of emotional responses (non-acceptance), difficulties engaging in goal-directed behavior (goals), impulse control difficulties (impulse), lack of emotional awareness (awareness), limited access to emotion regulation strategies (strategies), lack of emotional clarity (clarity). Each item is rated on a 5-point Likert scale (from 1 = almost never to 5 = almost always), giving a total score range of 36 to 180.
- Adult Rejection Sensitivity Questionnaire (A-RSQ) for rejection sensitivity [44]. Rejection sensitivity (RS) refers to a cognitive-affective disposition to anxiously expect rejection, shaped by one's cognitive-social learning history and activated in situations involving potential rejection or acceptance. The A-RSQ is an adaptation of the RSQ [45]. It includes 9 items describing scenarios in which one might ask others for something. For each scenario, participants respond to: (a) their concern about being rejected, and (b) their expectation of being accepted. Both questions are rated on a 6-point Likert scale (1 = very unconcerned/unlikely to 6 = very concerned/likely). The rejection expectancy is determined by subtracting the acceptance expectancy score from 7. The RS score for each scenario is the product of rejection concern and rejection expectancy. The total A-RSQ score is the average of the RS scores across all nine scenarios, resulting in a range from 1 to 36, with higher scores indicating greater rejection sensitivity.

The Adverse Childhood Experiences (ACE) Questionnaire [46,47], the Relationship Scale Questionnaire (RSQ) for attachment style (secure, fearful, preoccupied, dismissing) [48], the Montreal Cognitive Assessment (MoCA) [49] and Posttraumatic Stress Disorder (PTSD) Checklist for DSM-5 (PCL-5) [50] were assessed only at baseline. The 5-Dimensional Altered States of Consciousness Scale (5D-ASC) [51] was administered after the end of subjective acute effects at hour 6 of dosing. This scale encompasses 11 dimensions: oceanic boundlessness, which includes disembodiment, insightfulness, blissful state, spiritual experience and experience of unity; dread of ego dissolution, which includes impaired control and cognition and anxiety; and visionary restructuralization, which includes changed meaning of percepts, audio-visual synesthesia, elementary imagery and complex imagery [52].

Bias was reduced by collecting questionnaire data on-line without interaction with the clinical team. Additionally, drinking outcomes were collected using the TLFB by an independent and blinded assessor.

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Statistical analysis

Sample size

For this pilot study, the number of subjects required was determined according to the recommendations of Hertzog [53] (i.e. 20 patients in the experimental group and 10 patients in the control group).

Results for the primary and other endpoints are presented using a per-protocol approach, as treated (n = 20:10). Quantitative variables are presented as median and interquartile range or mean and SD. Qualitative variables are presented as numbers and percentage. The rate of patients completing both dosing sessions (primary endpoint) was compared between groups via a Fischer test. For secondary outcomes, qualitative variables were compared between groups using a χ^2 or Fisher's exact test. Quantitative variables were compared between groups using a Student's t test if the variable had a Gaussian distribution, or a non-parametric Mann–Whitney test otherwise. We reported the description and comparison of baseline characteristics and dimensions of the psychedelic experience (5D-ASC) in the subgroups of abstainers and non-abstainers at week 12 in the 25 mg group to explore for response factors. We compared the role selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) on drinking outcomes. We reported a survival analysis of the time to relapse as defined as first HDD. Time to relapse was explored using Cox regression model to estimate hazard ratio and a survival curve was provided. An exploratory linear mixed model compared the evolution of the DERS total score between the two groups. All analyses were performed using R software, version 4.4.0. Significance was set to the $\alpha = 0.05$ threshold. Because only 1 week 12 endpoint was missing, and efficacy analyses were secondary outcomes, we simply excluded this observation in the analyses except in the survival analysis where relapse had already occurred at week 6. We provide a sensitivity analysis using the last observation carried forward (LOCF) approach in the Supporting information.

RESULTS

Population

During the recruitment period (5 February 2024 to 24 October 2024), 350 potential participants were screened, of whom 30 were definitively included and randomized. The per protocol population was 20 participants in the 25 mg psilocybin group and 10 in the 1 mg psilocybin group. We were unable to collect screening delays because of a multi-step email-based process and participant-initiated screening withdrawals. Most requests to participate were spontaneous (98%) (Figure 2), and common reasons for exclusion were absence of AUD mostly people with depression only or with another condition only, e.g. obsessive-compulsive disorder) (23%), and co-addictions (12%). Mean participant age was 49 ± 10, and 43% were female. The mean number of HDD in the last drinking period at inclusion was 20 ± 9, with 21 ± 9 drinking days and a daily alcohol consumption of 12.9 ± 9.8 drinks. Besides depression, 70% of participants presented another comorbid mental health disorder, including, 23% with PTSD as assessed with the Mini International Neuropsychiatric Interview. At inclusion, 10% of controls and 55% in the 25 mg group were on antidepressant treatment, without discontinuation, while 20% of controls

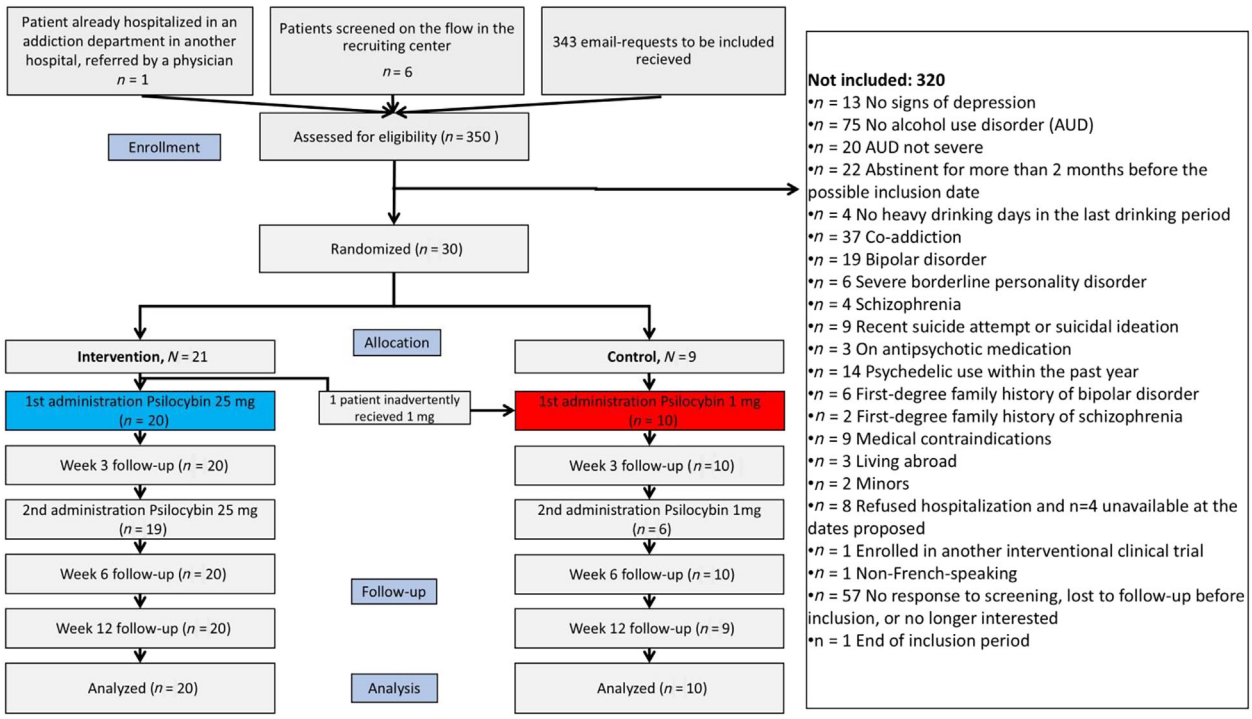


FIGURE 2 Flow chart. *Note:* Turquoise blue box is the per-protocol 25-mg group, and the red box is the per-protocol 1-mg group.

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and 10% in the 25 mg group were on antidepressant treatment, but discontinued it. Table 1 presents baseline participants characteristics. Table S1 presents co-medications.

Feasibility

One participant in the 25 mg group did not receive the second dose because of a myocardial infarction occurring 3 days before the second dosing session, attributed to undiagnosed coronary artery disease. Additionally, four participants in the control group refused the second dose, having suspected their treatment group. Three of them left the facility before the end of the relapse prevention program. Three declared their intention to self-administer psychedelics after discharge, with one already having taken 3,4-methylenedioxymethamphetamine once during follow-up. Participants in the experimental group had a higher participation rate in the second dosing session (95% in the 25 mg group vs. 60% in the control group, *P*-value = 0.031). All included patients received the first dosing session and the integration visit.

Acute effects (oral 0–10 scale)

The maximum experience intensity was 10 in the 25 mg group and 6 in controls. Two patients in the 25 mg group reported anxiety level 10, which decreased to 0 in 30 minutes, whereas the maximum level of anxiety in controls was 4 (Figure 3). The maximum perceived level of threat felt by a participant in the 25 mg group was 5, decreasing to 0 in 30 minutes. No rescue treatments were necessary.

Radar plots of the 5D-ASC questionnaire showed similar shape and intensity of the experience for both dosing sessions in the 25 mg group, except anxiety was lower in the second session (Figure 4).

Role of antidepressant in the 25 mg group

Figure 5 presents the 5D-ASC radar plots according to antidepressant use in the 25 mg group. In both sessions, all dimensions were slightly diminished on antidepressants, especially audio-visual synesthesia and anxiety. The only significant difference was for anxiety during the first dosing session [34 (17; 49) vs. 2 (1; 21) $P = 0.021$], and elementary imagery and audio-visual synesthesia during the second session [96 (84; 98) vs. 54 (19; 86), $P = 0.048$ and 96 (87; 98) vs. 46 (17; 90), $P = 0.048$]. Table S2 presents session 1 and 2 dimensions of the 5D-ASC with and without antidepressants at baseline and their comparison.

Preliminary efficacy on alcohol-related outcomes and depression

Table 2 presents the evolution in alcohol-related and depression outcomes. At 12 weeks, we found a significant between group difference in abstinent rate [11/20 (55%) vs. 1/9 (11%) (one lost of follow up) (difference = −44%, [95% CI = −82%, −5.9%], $P = 0.043$), reductions in % drinking days −100 (−100; −49) vs. − 93 (−96; 0), $P = 0.038$ and craving frequency −8 (−23; −1) vs. + 7 (−2; 11), $P = 0.045$, respectively, in the 25 vs. 1 mg groups (median [25;75])]. In addition, 45% of participants in the 25 mg group were classed as responders at week 12 in terms of 80% reduction of BDI score compared to none in the control group (*P*-value = 0.027). Survival analysis for the event first HDD (= relapse): relapse rates were 50% in the control group and 35% in the 25 mg group. The hazard ratio for relapse was 0.52 (0.16; 1.65), demonstrating a non-significant reduction of relapse in heavy drinking. Figure 6 shows the survival curve. In the subsample of non-abstinent at week 12 participants, the median number of drinking days was 3 (1; 13) in the 1 mg group ($n = 9$) and 7 (1; 19) in the 25 mg group ($n = 9$), with no significant difference between groups ($P = 0.9$). This *post hoc* analysis should be interpreted with caution, as excluding abstinent participants may introduce negative bias, particularly given the significantly higher proportion of abstainers in the 25 mg group at week 12. Sensitivity analysis using the LOCF approach is presented in Supplementary Table S3. Individual efficacy outcomes are presented in Supplementary Table S4.

Evolution of emotion regulation

Figure 7 presents an exploratory linear mixed model comparing the evolution of the DERS total score between the two groups. The observed difference was not significant.

Role of antidepressants on drinking outcomes

At inclusion, participants in the 25 mg group taking an antidepressant ($n = 13$) drank on median 192 (96; 448) standard drinks in the last 4-week drinking period, compared to 569 standard drinks (409; 773) in those without an antidepressant ($n = 7$), a difference that did not reach significance (*P*-value = 0.068). The median number of drinking days did not differ [23 (10; 28) with antidepressant versus 28 (26; 28) without, *P*-value = 0.2], nor did the number of HDD [22 (10; 28) versus 28 (25; 28), *P*-value = 0.3]. We did not find any difference in efficacy outcomes in the 25 mg group according to antidepressant use at inclusion (Table 3).

Factors associated with response as assessed by abstinence at week 12

Baseline PCL-5 score was significantly associated with response as assessed by abstinence [median 39 (22; 45) in the non-responder group versus 48 (46; 53) in the responder group, *P*-value = 0.036]. All other baseline characteristics and acute psychedelic experience dimension demonstrated no significant association with abstinence. Table S5 presents comparison of baseline characteristics and dimensions of the psychedelic experience (5D-ASC) at session 1 of abstainers and non-abstainers at week 12 in the 25 mg group.

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TABLE 1 Baseline participant characteristics.

Baseline characteristics	Overall *n* = 30	Psilocybin 1 mg *n* = 10	Psilocybin 25 mg *n* = 20
Age (y)	50 (42; 57)	51 (45; 57)	49 (41; 57)
Sex (female)	13 (43%)	4 (40%)	9 (45%)
Body mass index (kg/m^2)	24.3 (22.1; 28.4)	24.0 (22.2; 25.2)	27.1 (22.1; 28.7)
Adverse Childhood Events Questionnaire	5.00 (3.0; 6.0)	5.0 (2.3; 7.5)	5.00 (3.0; 6.0)
Current mental health disorder (MINI), *n* (%)			
Panic disorder	0 (0)	0 (0)	0 (0%)
Agoraphobia	1 (3.3)	0 (0)	1 (5.0%)
Social phobia	3 (10)	2 (20)	1 (5.3%)
Obsessive compulsive disorder	1 (3.3)	0 (0)	1 (5.0%)
Posttraumatic disorder	7 (23)	3 (30)	4 (20%)
Anorexia	2 (6.7)	1 (10)	1 (5.0%)
Bulimia	1 (3.3)	0 (0)	1 (5.0%)
Binge eating disorder	4 (13.3)	2 (20)	2 (10%)
General anxiety disorder	16 (53.3)	7 (70)	9 (45%)
On antidepressant	17 (56.7%)	3 (30%)	14 (70%)
MoCA	28 (26; 29)	29 (26; 29)	28 (27; 29)
PCL5	45 (31; 51)	43 (32; 50)	46 (32; 50)
RSQ: secure	2.6 (2.4; 3.0)	2.6 (2.5; 2.8)	2.7 (2.4; 3.2)
RSQ: fearful	3.25 (2.6; 3.9)	3.38 (3.3; 3.9)	3.13 (2.5; 3.9)
RSQ: preoccupied	3.25 (2.8; 3.8)	3.5 (3.1; 3.9)	3.0 (2.6; 3.6)
RSQ: dismissing	3.60 (3.0; 4.0)	3.40 (3.1; 3.8)	3.7 (2.9; 4.0)
Beck anxiety inventory	15 (9; 20)	14 (8; 20)	16 (9; 22)
DERS: total score	116 (101; 132)	113 (96; 119)	119 (109; 132)
DERS: non-acceptation	19 (16; 22)	19 (16; 20)	19 (16; 23)
DERS: goal	19.0 (16.0; 20.8)	17.0 (15.3; 20.3)	20.0 (17.5; 20.5)
DERS: impulsivity	17.0 (14.0; 21.8)	15.0 (12.5; 17.8)	17.5 (14.0; 23.0)
DERS: aware	20.0 (15.0; 25.0)	17.5 (15.0; 21.5)	21.5 (14.5; 27.0)
DERS: strategies	28.5 (23.3; 32.0)	25.0 (19.3; 29.3)	30.0 (24.0; 33.0)
DERS: clarity	15.5 (13.0; 18.0)	14.0 (13.0; 16.0)	16.0 (13.0; 18.0)
MLQ total score	42 (36; 48)	47 (41; 50)	39 (35; 46)
MLQ: presence	15 (12; 22)	18 (15; 21)	15 (11; 22)
MLQ: research	27 (23; 32)	29 (26; 32)	26 (22; 32)
Beck Depression Inventory	34 (24; 39)	33 (24; 36)	36 (25; 40)
Adult Rejection Sensitivity Questionnaire	10.6 (8.3; 14.2)	8.7 (7.2; 11.3)	11.8 (8.7; 15.8)
CEQ-intensity	61 (52; 76)	62 (50; 73)	60 (55; 77)
CEQ-frequency	20 (8; 51)	15 (8; 26)	21 (8; 56)
Alcohol Quality of Life Scale	53 (32; 67)	45 (32; 51)	58 (39; 69)
Alcohol consumption in the last 4-week drinking period			
No. of heavy drinking days	23 (13; 28)	21 (16; 24)	28 (12; 28)
Total alcohol consumption (10 g drinks)	306 (163; 477)	256 (191; 406)	345 (126; 605)
No. of drinking days	24 (13; 28)	23 (16; 27)	28 (12; 28)

Note: Data are median (25%; 75%) or *n* (%).

Abbreviations: CEQ, Craving Experience Questionnaire; DSM-5, Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition; MINI, Mini International Neuropsychiatric Interview; MoCA, Montreal Cognitive Assessment; PCL, Posttraumatic Stress Disorder Checklist for DSM-5; RSQ, Relationship Scale Questionnaire; DERS, Dysfunctional Emotion Regulation Scale; MLQ, Meaning in Life Questionnaire.

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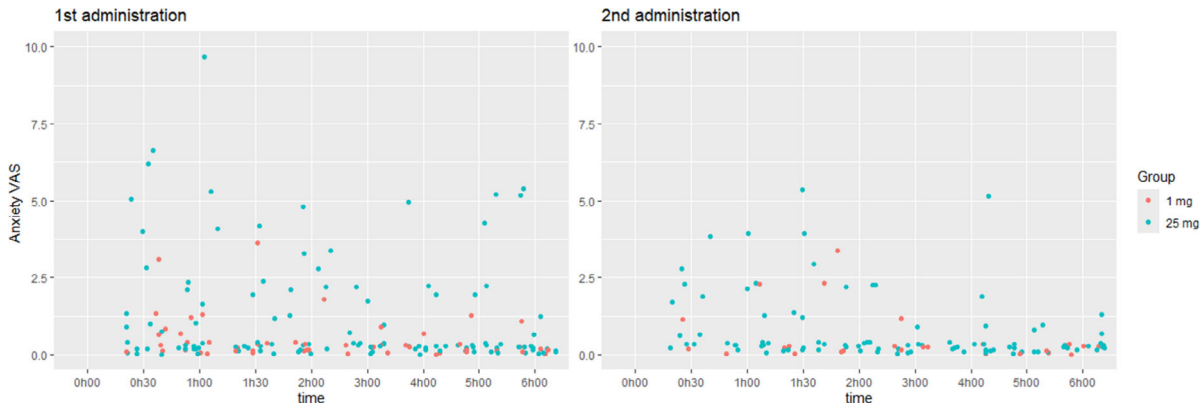


FIGURE 3 Dot plots of anxiety levels over the 6-hour dosing sessions by groups (0–10 VAS).

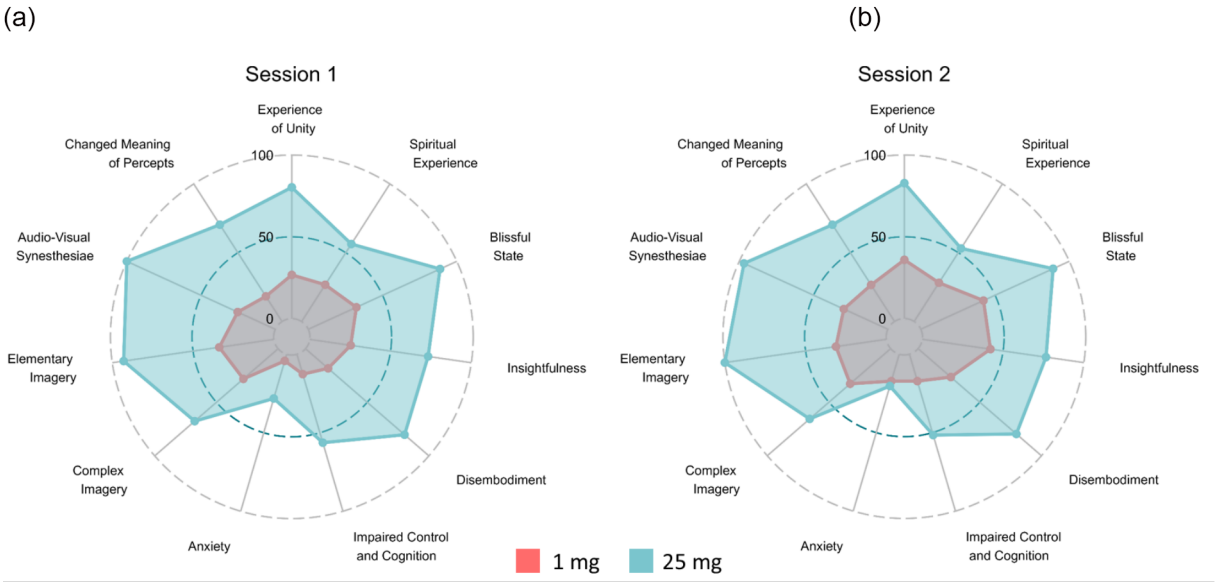


FIGURE 4 Radar plots of 5-Dimensional Altered States of Consciousness Scale (5D-ASC) dimensions for the first (a) and second (b) dosing session.

Safety

Nineteen adverse events were reported in 10 patients (50%) in the 25 mg group, compared to six adverse events in six patients (60%) in the control group (Table 4), with some patients experiencing multiple events. A total of four events (21%) in the 25 mg group and one event (17%) in the 1 mg group were considered attributable to the treatment. In the experimental group, we reported two episodes of spontaneously resolving dizziness in the same participant during the dosing session, and one patient reported nightmares beginning 2 days after the first dosing session and lasting 9 days. Finally, hypoglycemia occurred in a patient with comorbid anorexia at the end of the session and was resolved with a snack. In all cases where the event was not attributable to the treatment, it occurred either before the first dosing session or several days to weeks afterward, and was better explained by the natural course of the disease or another underlying health condition.

Suicidal risk

In the control group, one patient presented suicidal ideation related to suspecting their group assignment. Another patient attempted suicide 1 month after the first dosing session coinciding with alcohol relapse and encountering a stressful life event and subsequently refused the second dosing session. In the 25 mg group, two patients presented suicidal ideation during a heavy drinking occasion after discharge.

Blinding guess

Among patients, the correct blinding guess rate was 93.33%, compared to 86.67% in investigators (Table 5). Patients in the control group more frequently guessed their group correctly than the investigators, with the inverse true for the experimental group.

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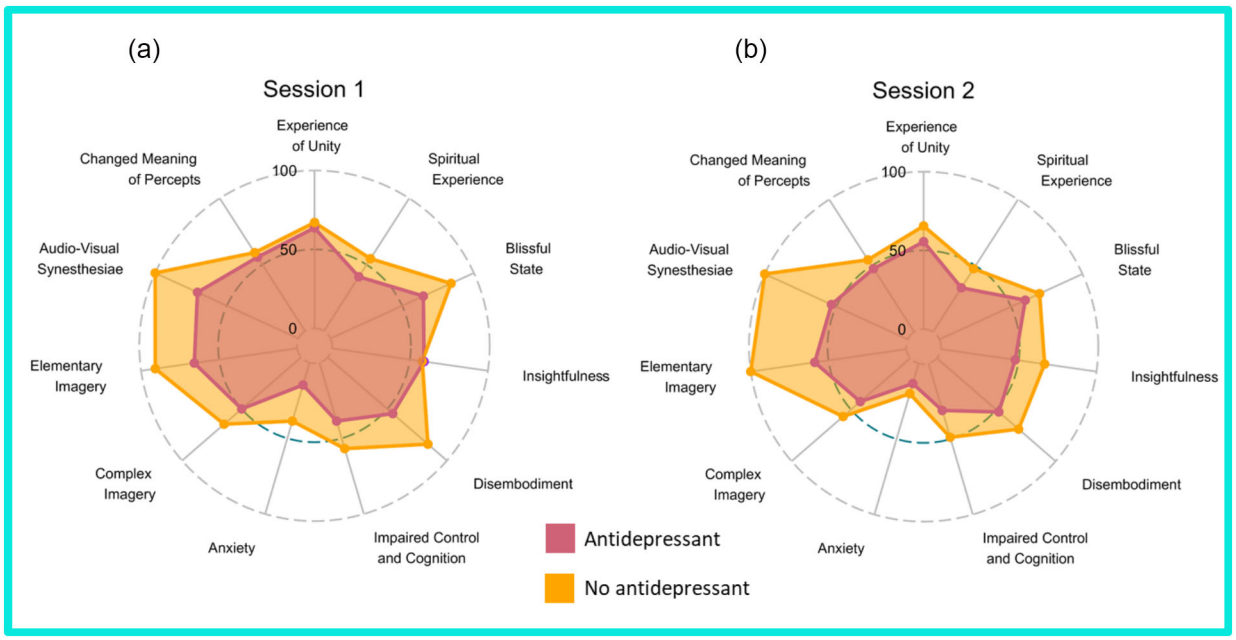


FIGURE 5 Radar plots of 5-Dimensional Altered States of Consciousness Scale (5D-ASC) dimensions according to current antidepressant use for the experimental group (25 mg) for the first (a) and second (b) dosing session.

DISCUSSION

Our pilot study, recruiting 30 participants at a single center in less than 9 months, demonstrated the feasibility of, and high demand for, psychedelic-assisted psychotherapy for dual disorder. Moreover, our results support the acceptability and feasibility of the schedule for individuals with AUD and depressive symptoms as an add-on to treatment as usual. Anecdotally, participation in the study appeared to serve as a motivational lever for some patients to initiate detoxification and should be investigated in future research.

Although efficacy was not the primary outcome and despite our limited sample size, we observed significant differences favoring the 25 mg group for several AUD and depression endpoints. The abstinence rate was significantly five times higher in the 25 mg groups than in controls. Moreover, the 12-week percentage change in drinking days, craving frequency and the 12-week reduction of at least 80% of BDI score were significantly higher. Our results may provide important insights into positioning psilocybin-assisted psychotherapy (PAP) as a potential treatment for both the maintenance of abstinence and the reduction of alcohol consumption. Future studies should refine drinking outcome measures to better capture PAP effect and identify those most susceptible in terms of drinking goal. The other efficacy outcomes did not reach significance. Interestingly, no significant difference was found on the criteria reduction of at least 50% of BDI score, which is the one usually used in the literature [41]. This aligns with clinical observations that, in terms of depressive symptoms, responders often experience profound changes, with near-complete resolution of symptoms—an outcome that appears more difficult to achieve with other therapeutic strategies. The selection of meaningful and relevant outcomes is, therefore, critical, as traditional measures may not always capture the full therapeutic potential of psychedelics within this emerging paradigm. Interestingly, we found a significantly higher baseline PCL-5 in abstainers versus non-abstainers.

Anxiety levels were low, showing that one preparation meeting before each dosing session was sufficient. Although several interpretations could be done, the approach standardized in our therapeutic manual appears supported by these feasibility results. The manual emphasizes exposure to a modified state of consciousness through mindfulness practice during the preparation session, psychoeducation and the development of a horizontal, patient-centered therapeutic relationship. Mindfulness and ACT have been used in previous trials for depression [54]. The very low anxiety levels may be partly explained by the high proportion of patients taking antidepressants, which could have attenuated the intensity of the psychedelic experience [55]. However, our findings support recent studies suggesting that SSRIs and SNRIs do not reduce the therapeutic response to psychedelics or affect side effects [56]. Previous reports of increased side effects or reduced efficacy in self-selected samples may reflect set parameters, such as history of bipolar disorder, schizophrenia, severe borderline personality disorder or concurrent use of other substances [57]. Additionally, our decision to allow continued use of antidepressants was based on evidence indicating lower response rates among individuals who discontinued them before treatment [58]. Indeed, the 25 mg group showed maximal intensity of the psychedelic experience in participants taking SSRIs and SNRIs, with some dimensions of the experience, particularly for dimensions related to anxiety, appearing attenuated. However, this did not seem to affect overall treatment response, although the lake of significant difference in efficacy outcomes between the subgroups with and without

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TABLE 2 Alcohol and depression outcomes.

Efficacy outcomes Median (25%; 75%) or *n* (%)	Psilocybin 1 mg *n* = 10 at week 6 and *n* = 9 at week 12	Psilocybin 25 mg *n* = 20	*P*-value
Alcohol-related outcomes			
6-week abstinence	4 (40)	14 (70)	0.14
12-week abstinence	1 (11)	11 (55)	0.043*
6-week heavy drinking days	0.0 (0.0; 0.0)	0.00 (0.0; 0.0)	0.7
12-week heavy drinking days	0 (0; 2)	0 (0; 1.3)	>0.9
6-week % change in heavy drinking days	−100 (−100; −100)	−100 (−100; −100)	>0.9
12-week % change in heavy drinking days	−100 (−100; −88)	−100 (−100; −95)	>0.9
6-week drinking days	1.5 (0.0; 4.5)	0.00 (0.0; 1.3)	0.14
12-week drinking days	2 (1; 13)	0 (0; 3)	0.065
6-week % change in drinking days	−93 (−100; −66)	−100 (−100; −96)	0.10
12-week % change in drinking days	−93 (−96; 0)	−100 (−100; −49)	0.038*
6-week total alcohol consumption	4 (0; 22)	0 (0; 6)	0.14
12-week total alcohol consumption	3 (1; 40)	0 (0; 52)	0.3
6-week % change in total alcohol consumption	−98 (−100; −85)	−100 (−100; −99)	0.12
12-week % change in total alcohol consumption	−99 (−100; −80)	−100 (−100; −88)	0.2
12-week Alcohol Quality of Life scale total score	19 (9; 25)	14 (4; 21)	0.3
12-week change in Alcohol Quality of Life scale total score	24 (18; 27)	40 (14; 63)	0.2
12-week Craving Experience Questionnaire-intensity	31 (17; 47)	30 (9; 48)	0.7
12-week change in Craving Experience Questionnaire-intensity	−18 (−32; −16)	−39 (−52; −22)	0.14
12-week Craving Experience Questionnaire-frequency	22 (11; 40)	18 (1; 40)	0.3
12-week change in Craving Experience Questionnaire-frequency	7 (−2; 11)	−8 (−23; −1)	0.045*
Depression-related outcomes			
3-week Beck Depression Inventory	15 (11; 19)	13 (5; 19)	>0.9
6-week Beck Depression Inventory	15 (13; 18)	6 (2; 21)	0.15
12-week Beck Depression Inventory	13 (8; 20)	10 (3; 23)	0.4
3-week % change in Beck Depression Inventory	−48 (−64; −35)	−54 (−84; −31)	0.8
6-week % change in Beck Depression Inventory	−55 (−61; −22)	−81 (−96; −46)	0.090
12-week % change in Beck Depression Inventory	−60 (−63; −24)	−61 (−88; −27)	0.4
12-week 50% Beck Depression Inventory reduction	6 (67)	10 (50)	0.5
12-week 80% Beck Depression Inventory reduction	0 (0)	9 (45)	0.027*

*P-value <0.05.

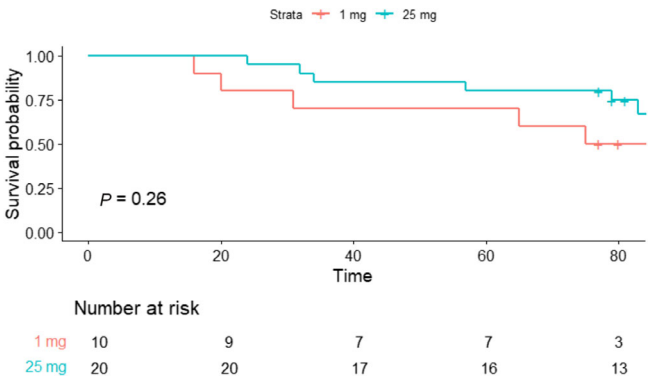


FIGURE 6 Survival curve for the event first heavy drinking day in the two groups.

antidepressant could be because of a lack of power. We reported the timing of antidepressant discontinuation; however, the reasons for discontinuation were not collected, which limits interpretability. Future studies could explore this dimension more systematically to better understand its clinical significance.

Our study had several strengths. The sample was well-balanced between male and female participants. Our sample included 43% women, which is higher than typically observed in studies involving patients with severe AUD in in-patient settings. For instance, a recent French cohort study [11] reported only 27% women in such contexts. Similarly, the two other randomized controlled trials (RCT) using psilocybin for AUD [12, 13], women comprised, respectively, 38% and 44% of the sample. These findings, including ours, may reflect a particularly high demand for alternative treatments—such as psychedelic-

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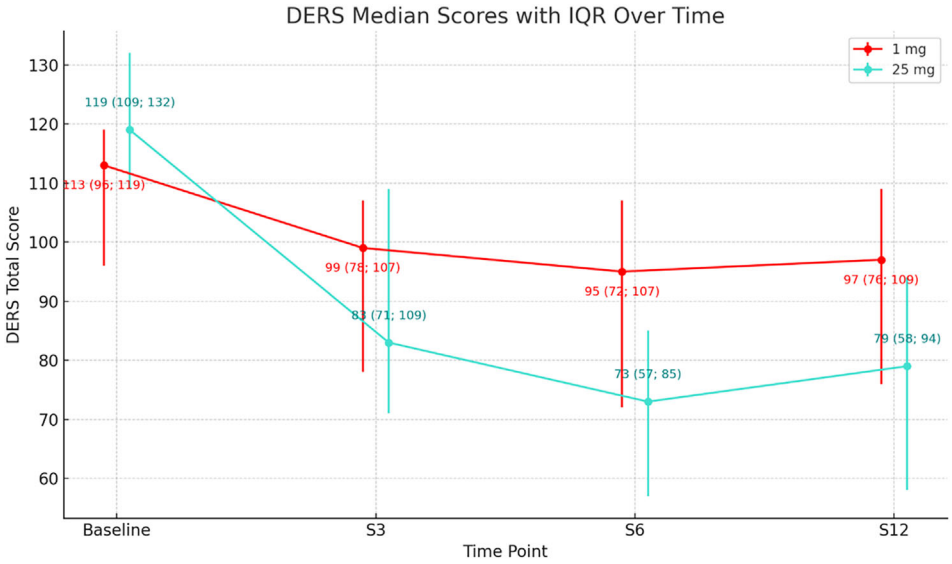


FIGURE 7 Evolution of the Difficulties in Emotion Regulation Scale (DERS) total score, median (25%;75%) in the two groups.

TABLE 3 Alcohol-related efficacy outcomes in the 25 mg group with and without antidepressant at inclusion.

Outcome	Subgroup without an antidepressant *n* = 7	Subgroup with an antidepressant *n* = 13	*P*-value
6-week abstinence	4 (57%)	10 (77%)	0.6
12-week abstinence	3 (43%)	8 (62%)	0.6
6-week heavy drinking days	0.0 (0.0; 0.0)	0.0 (0.0; 0.0)	0.9
12-week heavy drinking days	0.0 (0.0; 1.0)	0.0 (0.0; 2.0)	>0.9
6-week % change in heavy drinking days	−100 (−100; −100)	−100 (−100; −100)	0.9
12-week % change in heavy drinking days	−100 (−100; −96)	−100 (−100; −54)	>0.9
6-week drinking days	0.00 (0.0; 3.0)	0.0 (0.0; 0.0)	0.5
12-week drinking days	1 (0; 10)	0 (0; 2)	0.5
6-week % change in drinking days	−100 (−100; −89)	−100 (−100; −100)	0.5
12-week % change in drinking days	−96 (−100; −64)	−100 (−100; −54)	0.6
6-week total alcohol consumption	0 (0; 1)	0 (0; 0)	0.5
12-week total alcohol consumption	6 (0; 37)	0 (0; 48)	0.7
6-week % change in total alcohol consumption	−100 (−100; −98)	−100 (−100; −100)	0.5
12-week % change in total alcohol consumption	−0.98 (−1.00; −0.94)	−1.00 (−1.00; −0.53)	>0.9

assisted therapy—among women. Moreover, the under-representation of women in RCTs is a widespread issue across all indications [14], raising concerns about the generalizability and equity of evidence-based medicine. In this context, the relatively high proportion of women in our sample is encouraging and suggests that the proposed intervention is acceptable and feasible not only for men, who remain the majority, but also for women—who often pay the price of being systematically under-represented in research. Nearly a quarter of participants had no prior expectations or knowledge about psychedelics, limiting expectancy bias and related boosting effect of psilocybin [59]. Furthermore, the relapse rate in the control group was comparable to rates in previous studies [8], illustrating the absence of a nocebo effect. Our population with severe AUD and add-on approach to an intensive relapse prevention program completes the two randomized trials in populations with less severe AUD [20] and with milder psychotherapy.

Although some participants refused the second dose, the study had a low drop-out rate, possibly because of the high demand for participation, as found in a recent Swiss trial in AUD [22]. Offering a guaranteed high-dose session later could overcome the refusal rate for the second dose in the control group and lower the risk of arm contamination from self-administration of psychedelics.

There are also several limitations. This was a pilot study in a single center with a small sample size, limiting the representativeness of

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TABLE 4 List of adverse events.

Adverse events, *n* (%)	1 mg group *n* = 10		25 mg group *n* = 20	
	Attributable to the treatment	Not attributable to the treatment	Attributable to the treatment	Not attributable to the treatment
Dizziness			2 (10)	
Headache				2 (10)
Diarrhea				1 (5)
Nausea				1 (5)
Gastroenteritis				1 (5)
Rumination		1 (10)		
Nightmares			1 (5)	
Fatigue			1 (5) (attributability NA)	
Suicidal ideation	1 (10)			2 (10)
Suicidal attempt		1 (10)		
Hospitalization for relapse in AUD				2 (10)
Ocular hypertension				1 (5)
Myocardial infarction				1 (5)
Ankle sprain		1 (10)		
Hypoglycemia			1 (5)	
Discovery of a breast lump on entry examination before treatment				1 (5)
Cough				1 (5)
Palpebral abscess		1 (10)		
COVID-19/non-specific viral illness		1 (10)		1 (5)

Abbreviations: AUD, alcohol use disorder; COVID-19, coronavirus disease 2019.

TABLE 5 Patient and investigator blinding guess.

Blinding guess	1 mg group (*n* = 10)	25 mg group (*n* = 20)	*P*-value
Patient			<0.001*
1 mg	10 (100%)	2 (10%)	
25 mg	0 (0%)	18 (90%)	
Clinical investigator			<0.001*
1 mg	8 (80%)	1 (5.0%)	
25 mg	2 (20%)	19 (95%)	

Note: Grey shading, correct blinding guess.
**P*-value <0.05.

the sample. The cohort with severe AUD and depression may not reflect the broader AUD population. We were unable to collect all feasibility outcome measures (screening delays) because of a multi-step email-based process and participant-initiated screening withdrawals, which were documented only in terms of frequency. However, unlike most studies, we documented the reason for non-inclusion for every inclusion request, providing rare and valuable insight into the unmet needs of individuals living with mental disorders who saw psilocybin-assisted psychotherapy as a potential treatment option.

As recommended [60], we reported blinding integrity. Although blinding integrity is often criticized in psychedelic studies, successful blinding is not an issue specific to psychedelics, being below 50% across all published clinical trials [61]. In depression trials, only 7% report blinding integrity [60]. Our patients' blinding guess rate aligns with previous findings in AUD, which range from 91% to 95% [20, 22]. Clinical investigators correctly guessed group attribution less often than in the two other randomized studies in AUD that used active placebo, supporting very low dose psilocybin as a control. Using a very low dose as a comparator rather than another active

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placebo aligns with United States Food and Drug Administration and European Medicines Agency recommendations [62, 63]. It also avoided induced side effects, minimizing underestimation of the safety of the psychedelic compound [60]. Future studies might consider a 3 mg control dose, although initial efficacy signals have been reported at moderate doses (10 mg) in depression [64].

Given the role of expectations, which can change after an initial dosing session [60], the preparation sessions focused on neutralizing pre-existing expectations. This included warnings about the non-constant nature of hallucinations, variability in subjective effects between dosing sessions and the importance of the treatment as an add-on rather than a standalone intervention. These measures may have helped mitigate any 'lessebo' effect [60], given that relapse rates in the control group were not higher than typically reported in this population.

The further reduction in depressive symptoms after the second dose supports the schedule of two sessions spaced 3 weeks apart. Some participants experienced varying subjective effects between sessions, not necessarily indicating tolerance but, in some cases, reflecting increased intensity of the psychedelic experience. The second dosing session allowed some participants to fully engage emotionally after initially reacting with euphoria, which could have served as an emotional avoidance strategy. Not giving a second dose could limit the effect size in patients with AUD [22]. Finally, the study used an unbalanced 2:1 randomization ratio to maximize data collection in the 25 mg group while also assessing the acceptability of the randomized design. Although this could theoretically enhance the placebo effect because of the increased probability of receiving 25 mg [60], this risk was likely limited, as participants guessed their allocation at a lower rate than clinical investigators in the 25 mg group.

CONCLUSION

This study provides promising feasibility and preliminary efficacy data on the use of two 25 mg psilocybin-assisted psychotherapy sessions, spaced 3 weeks apart, as an add-on to an intensive relapse prevention program for patients with alcohol use disorder and co-occurring depression. High participant demand, low anxiety levels during sessions and the ability to continue antidepressant treatment support the acceptability of this approach. The findings highlight the potential therapeutic value of psilocybin in this dual-diagnosis population. Mechanistic investigations and a larger efficacy trial are warranted. For future studies, we recommend increasing the control dose to 3 mg. Additionally, offering a third high-dose session to all participants could help assess the utility of a maintenance dose and reduce contamination bias by mitigating the risk of self-administration in disappointed control group participants.

AUTHOR CONTRIBUTIONS

Conceptualization: Amandine Luquiens and Thibault Mura. *Data curation*: Chris Serrand and Thibault Mura. *Formal analysis*: Chris Serrand and Thibault Mura. *Funding acquisition*: Amandine Luquiens and Thibault Mura. *Investigation and resources*: Amandine Luquiens, Dahbia Belahda, Carine Graux, Noe Igounenc, Paul Rochefort and Felix Sergent. *Methodology and software*: Amandine Luquiens, Chris Serrand and Thibault Mura. *Project administration*: Amandine Luquiens and Thibault Mura. *Supervision*: Amandine Luquiens and Thibault Mura. *Validation*: Amandine Luquiens and Thibault Mura. *Visualization*: Amandine Luquiens, Chris Serrand and Thibault Mura. *Writing—original draft*: Amandine Luquiens, Chris Serrand and Thibault Mura. *Writing—review and editing*: Amandine Luquiens, Dahbia Belahda, Carine Graux, Noe Igounenc, Paul Rochefort, Felix Sergent, Chris Serrand and Thibault Mura.

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DECLARATION OF INTERESTS

None.

DATA AVAILABILITY STATEMENT

In accordance with the French Data Protection Act (Loi Informatique et Libertés) and the General Data Protection Regulation (RGPD), individuals are entitled to access, rectify and request the deletion of their personal data and to exercise their right to restrict the processing of their data. They may withdraw their consent to the processing of their data at any time; they may also object to the processing of their data; and they must be informed of any new processing outside the original purpose for which the data was collected. This very strict European framework does not allow us to put this data on an open site, but de-identified data are available upon reasonable request from the corresponding author.

CLINICAL TRIAL REGISTRATION

This study was registered on clinicaltrials.gov (NCT06235411).

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